Peterson's Algorithm

- Peterson's Algorithm is a software solution to the Critical Section Problem for two processes.
- It was proposed by Gary **Peterson** (1981) and is considered an improved version of the Turn Variable solution.

```
while (true) {

  flag[i] = true;     // I want to enter
  turn = j;           // Give priority to other process

  while (flag[j] && turn == j);

              // Critical Section

  flag[i] = false;

  // Remainder Section
}
```

Mutual Exclusion - Yes
Progress - Yes
Bounded Waiting - Yes

Banker’s Algorithm

- Multiple instances of resources
- Each process must a priori claim maximum use
- When a process requests a resource, it may have to wait
- When a process gets all its resources it must return them in a finite amount of time

Data Structures for the Banker's Algorithm

Let n = number of processes, and m = number of resources types.

- **Available***:* Vector of length m. If available [j] = k, there are k instances of resource type R_j available
- **Max***: n x m* matrix. If *Max* [i,j] = k, then process P_i may request at most k instances of resource type R_j
- **Allocation***:* n x m matrix. If Allocation[i,j] = k then P_i is currently allocated k instances of R_j
- **Need***:* n x m matrix. If *Need*[i,j] = k, then P_i may need k more instances of R_j to complete its task

Need [i,j] = *Max*[i,j] – *Allocation* [i,j]

Safety Algorithm / Banker's Algorithm

1. Let ***Work*** and ***Finish*** be vectors of length *m* and *n*, respectively. Initialize:

   Work* = *Available

   ***Finish* [*i*] = *false* for *i* = 0, 1, …, *n*- 1**

2. Find an ***i*** such that both:

   (a) ***Finish* [*i*] = *false***

   (b) $\boldsymbol{Need_i \leq Work}$

   If no such ***i*** exists, go to step 4

3. $\boldsymbol{Work = Work + Allocation_i}$

   Finish*[*i*] = *true

   go to step 2

4. If ***Finish* [*i*] == *true*** for all ***i***, then the system is in a safe state

Example of Banker's Algorithm

- 5 processes P_0 through P_4;

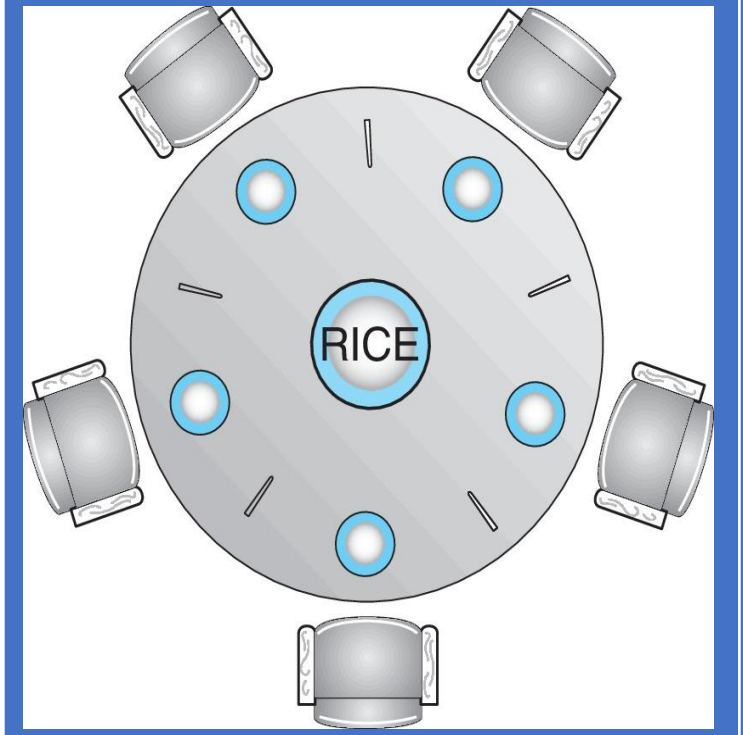
  3 resource types:

  A (10 instances), *B* (5 instances), and *C* (7 instances)

- Snapshot at time T_0:

	Allocation	*Max*	*Available*
	A B C	*A B C*	*A B C*
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

Solution of Dining Philosophers Problem using semaphore




```
#define N 5

void Philosopher(int i)
{
   while (true)
   {
      Thinking();
      take_fork(i);
      take_fork((i + 1) % N

      EAT();

      put_fork(i)
      put_fork((i + 1) % N);
   }
}
```

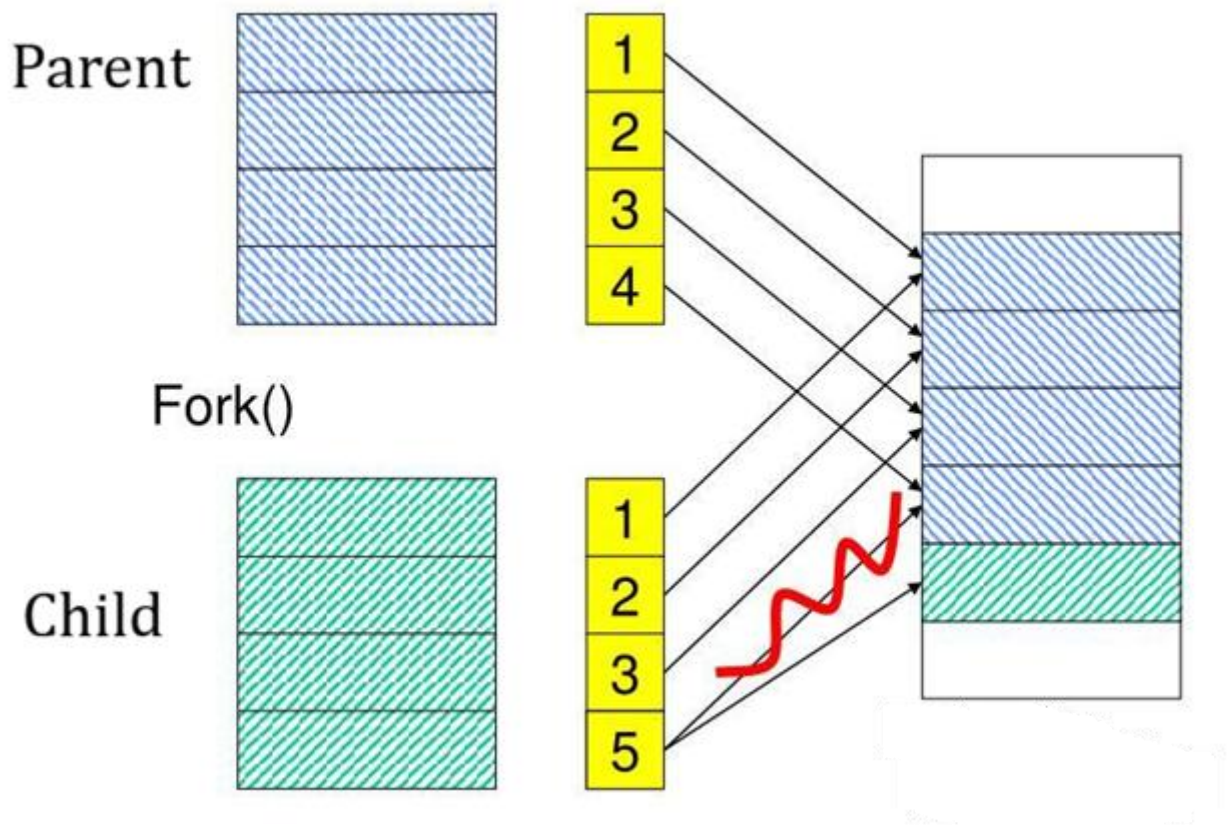
Philosopher	Left Fork	Right Fork
P0	S0	S1
P1	S1	S2
P2	S2	S3
P3	S3	S4
P4	S4	S0

Demand Paging

- Pages are loaded into memory only when needed.

- Efficient memory use
- Faster startup

Copy on write



Linux API

- fork() → create child
- exec() → load new program
- wait() → parent waits

C Program Forking Separate Process

```
#include <sys/types.h>
#include <stdio.h>
#include <unistd.h>

int main()
{
pid_t pid;

    /* fork a child process */
    pid = fork();

    if (pid < 0) { /* error occurred */
       fprintf(stderr, "Fork Failed");
       return 1;
    }
    else if (pid == 0) { /* child process */
       execlp("/bin/ls","ls",NULL);
    }
    else { /* parent process */
       /* parent will wait for the child to complete */
       wait(NULL);
       printf("Child Complete");
    }

    return 0;
}
```

IPC POSIX Producer

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

int main()
{
/* the size (in bytes) of shared memory object */
const int SIZE = 4096;
/* name of the shared memory object */
const char *name = "OS";
/* strings written to shared memory */
const char *message_0 = "Hello";
const char *message_1 = "World!";

/* shared memory file descriptor */
int shm_fd;
/* pointer to shared memory obect */
void *ptr;

    /* create the shared memory object */
    shm_fd = shm_open(name, O_CREAT | O_RDWR, 0666);

    /* configure the size of the shared memory object */
    ftruncate(shm_fd, SIZE);

    /* memory map the shared memory object */
    ptr = mmap(0, SIZE, PROT_WRITE, MAP_SHARED, shm_fd, 0);

    /* write to the shared memory object */
    sprintf(ptr,"%s",message_0);
    ptr += strlen(message_0);
    sprintf(ptr,"%s",message_1);
    ptr += strlen(message_1);

    return 0;
}
```

IPC POSIX Consumer

```
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <sys/shm.h>
#include <sys/stat.h>

int main()
{
/* the size (in bytes) of shared memory object */
const int SIZE = 4096;
/* name of the shared memory object */
const char *name = "OS";
/* shared memory file descriptor */
int shm_fd;
/* pointer to shared memory obect */
void *ptr;

    /* open the shared memory object */
    shm_fd = shm_open(name, O_RDONLY, 0666);

    /* memory map the shared memory object */
    ptr = mmap(0, SIZE, PROT_READ, MAP_SHARED, shm_fd, 0);

    /* read from the shared memory object */
    printf("%s",(char *)ptr);

    /* remove the shared memory object */
    shm_unlink(name);

    return 0;
}
```

NVM Scheduling

Feature	HDD	NVM (SSD/NVMe)
Movement	Mechanical (seek time)	No moving parts
Latency	High	Very low
Parallelism	Limited	High
Scheduling Need	High	Different approach

- Exploit parallelism
- Reduce write amplification
- Balance read/write

Types of NVM Scheduling Algorithms

- FIFO (First In First Out) - Simple
- Deadline Scheduling - Each request has a deadline
- NOOP Scheduler - Minimal reordering
- CFQ (Completely Fair Queuing) - Fair distribution among processes, time slicing
- Multi-Queue Scheduling (MQ) - Parallel request handling